

DESIGN AND ANALYSIS OF ALGORITHMS**(2019 Pattern) (Semester - VII) (410241)**

Time : 2½ Hours] [Max. Marks : 70

Instructions to the candidates:

1. Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.
4. Assume suitable data, if necessary.

Q1)

a) What is job scheduling algorithm? How job scheduling algorithm can be solved using Greedy algorithm approach? Explain your answer with respect to Principle, control abstraction, time analysis of control abstraction, of greedy approach for the following instance of Knapsack problem.

Each job is associated with a deadline and profit.

Job	J1	J2	J3	J4	J5
Deadline	2	1	3	2	1
Profit	60	100	20	40	20

[12]

b) Explain 0/1 knapsack problem using dynamic programming for following instance

Input: $N = 3$, $W = 4$

values[] = {1, 2, 3}

weight[] = {4, 5, 1}

[6]**OR**

a) What is Dynamic Programming? What is optimal binary search tree? How dynamic programming can be used to solve OBST for following instance.

Keys	10	20	30	40
Frequency	4	2	6	3

[10]

b) Compare between greedy method and dynamic programming with respect to:

- i) Feasibility
- ii) Optimality
- iii) Recursion
- iv) Memorization
- v) Time complexity

[8]

Q3)

a) Explain Backtracking algorithmic approach with respect to Principle, control abstraction, time analysis of control abstraction for 8-queen problem.

[8]

b) What is sum of subset problem? Solve sum of subset problem for following instance using backtracking approach.

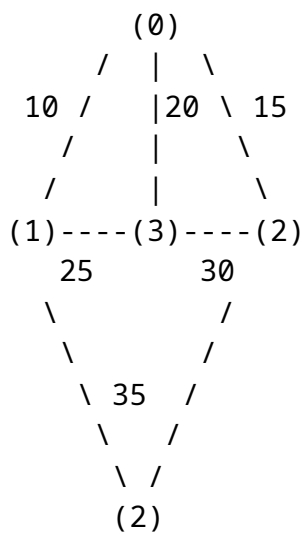
Input: $\text{set}[] = \{2, 3, 5, 6, 8, 10\}$, $\text{sum} = 10$

[9]

OR

Q4)

a) What is traveling salesman problem? Solve following figure for TSP using branch and bound method.



[9]

b) What is branch and bound algorithmic approach? Explain strategies — FIFO, LIFO and LC approaches for branch and bound method.

[8]

Q5)

a) Write short notes on the following:

- i) Aggregate Analysis
- ii) Accounting Method
- iii) Potential Function Method
- iv) Tractable and Non-tractable Problems

[10]

b) Write short notes on with suitable example of each:

- i) Randomized algorithm
- ii) Approximation algorithm

[8]

OR

Q6)

a) What is amortized analysis? Explain aggregate and potential function methods used for amortized analysis with respect to stack operations.

[9]

b) What is Potential function method of amortized analysis? To illustrate Potential method, find amortized cost of PUSH, POP and MULTIPOP stack operations.

[9]

Q7)

a) Write short notes on the following:

- i) Multithreaded matrix multiplication
- ii) Multithreaded merge sort
- iii) Distributed breadth first search
- iv) The Rabin-Karp algorithm

[10]

b) With respect to Multithreaded Algorithms explain:

- Analyzing multithreaded algorithms
- Parallel loops
- Race conditions

[7]

OR

Q8)

a) Write and explain pseudo code for Multi-threaded merge sort algorithm. How parallel merging gives a significant parallelism advantage over Merge Sort?

[9]

b) Write a pseudo code for naive string matching algorithm and Rabin-Karp algorithm for string matching and analyze the same.

[8]

[6627]-81